

2024, Mar. 2024 Release LOC:Acute Adult

Rhabdomyolysis or Crush Syndrome

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review ^(1, 2)Episode Day 1 ⁽¹⁾Episode Day 2 ⁽¹⁷⁾

Episode Day 3

Episode Day 4

Episode Day 5

Discharge Screens ⁽⁵³⁾

Utilization Benchmarks

Notes

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Overview

Instruction: This subset is intended to manage patients with rhabdomyolysis or crush syndrome. For patients with compartment syndrome please refer to the General Trauma subset.

Introduction:

Rhabdomyolysis is the degeneration (lysis) of skeletal muscle due to trauma, severe compression (e.g., compartment syndrome, crush syndrome), drugs, toxins, or infections. Lysis of muscle cells releases toxic intracellular components into the systemic circulation which may lead to electrolyte disturbances, hypovolemia, metabolic acidosis, coagulation defects, or acute kidney injury. Excessive physical exertion as well as direct muscle trauma can provoke rhabdomyolysis. Rhabdomyolysis is often observed in patients following traumatic injuries from motor vehicle accidents, industrial accidents, deep tissue burns, or severe physical abuse. Additionally, rhabdomyolysis may be observed in body-builders, marathon runners, individuals with substance use disorder, delirium tremens, or following protracted tonic-clonic seizures (Torres et al., The Ochsner Journal 2015, 15: 58-69).

Crush injury involves damage to striated muscle. Crush syndrome is the term used to describe complications resulting from muscle damage. Crush syndrome results in a characteristic syndrome of traumatic rhabdomyolysis, myoglobinuria, and acute renal failure. Renal and cardiac complications are dependent upon the magnitude of the pressure and the size of the compressed muscle group. Entrapment with muscle trauma can lead to compartment syndrome or rhabdomyolysis. Rhabdomyolysis under these conditions is potentially fatal (Gonzalez, Crit Care Med 2005; 33(1 Suppl): S34-41; Cervellin et al., Clin Chem Lab Med 2010, 48: 749-56). The Observation level of care is appropriate for patients with crush injury when there is concern for progression to a more severe state.

Evaluation and Treatment:

Patients with rhabdomyolysis often experience fever, fatigue, muscle pain, and localized edema. The initial clinical sign is the appearance of brown-red colored urine due to the presence of myoglobin. Diagnosis is based on laboratory findings, physical exam, and a thorough history. When rhabdomyolysis occurs, the need for continual monitoring of vital signs, urine output, arterial pH, electrolytes, creatine phosphokinase (CPK), cardiac status, and the avoidance of nephrotoxic agents is essential.

Early diagnosis and intervention is vital for a positive outcome. Early and vigorous fluid administration is essential for the prevention of acute kidney injury. Urine output must be carefully monitored as excessive fluid infusion can lead to congestive heart failure or pulmonary edema. Electrolyte imbalances must be monitored and corrected to avoid

potential cardiac arrhythmias. Prognosis depends on the extent and clinical severity of rhabdomyolysis. Most patients with mild renal impairment recover quickly, whereas others may require dialysis (Parekh et al., Emerg Med Pract 2012, 14: 1-15; Torres et al., The Ochsner Journal 2015, 15: 58-69).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:

● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽³⁾

- Rhabdomyolysis or crush syndrome ^(4, 5)
- Muscle edema or hematoma
- Myoglobinuria absent ⁽⁶⁾
- Prolonged compression lasting at least 1h ⁽⁷⁾
- Pulses decreased or absent

● **ACUTE, Both:** ⁽⁸⁾

- Rhabdomyolysis or crush syndrome ^(4, 5)

● **Finding, ≥ One:**

- Creatine phosphokinase (CPK) > 1,000 U/L(17.0 µkat/L) ⁽⁹⁾
- Myoglobinuria ⁽⁶⁾

● **CRITICAL, Both:** ⁽¹⁰⁾

- Rhabdomyolysis or crush syndrome ^(4, 5)

● **Finding, ≥ One:**

- Blood urea nitrogen (BUN) > 40 mg/dL(14.3 mmol/L)
- Calcium < 8 mg/dL(2.00 mmol/L)
- Creatinine > 2.0 mg/dL(176.8 µmol/L)

● **Electrocardiogram (ECG) changes, ≥ One:**

- Loss of P wave
- Multifocal PVCs
- Neuromuscular deficit ⁽¹¹⁾
- Widening QRS
- Potassium > 6.0 mEq/L(6.0 mmol/L)
- Urine output < 0.5 mL/kg/h

● **Episode Day 1, One:** ⁽¹⁾

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽³⁾

- Rhabdomyolysis or crush syndrome ^(4, 5)
- Muscle edema or hematoma
- Myoglobinuria absent ⁽⁶⁾
- Prolonged compression lasting at least 1h ⁽⁷⁾
- Pulses decreased or absent
- IV fluid ^(12, 13)

● **ACUTE, All:** ⁽⁸⁾

- Rhabdomyolysis or crush syndrome ^(4, 5)

● **Finding, ≥ One:**

- Creatine phosphokinase (CPK) > 1,000 U/L(17.0 µkat/L) ⁽⁹⁾
- Myoglobinuria ⁽⁶⁾

● **Intervention, All:**

- Electrolyte monitoring
- Renal function monitoring
- IV fluid ^(12, 13)

● **CRITICAL, All:** ⁽¹⁰⁾

- Rhabdomyolysis or crush syndrome ^(4, 5)

● **Finding, ≥ One:**

- Blood urea nitrogen (BUN) > 40 mg/dL(14.3 mmol/L)
- Calcium < 8 mg/dL(2.00 mmol/L)
- Creatinine > 2.0 mg/dL(176.8 µmol/L)

- **Electrocardiogram (ECG) changes, ≥ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹¹⁾
 - Widening QRS
 - Potassium > 6.0 mEq/L(6.0 mmol/L)
 - Urine output < 0.5 mL/kg/h
- **Intervention, ≥ One:**
 - Calcium chloride or gluconate
- **Dialysis, One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - CRRT ⁽¹⁴⁾
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
 - IV fluid resuscitation ≤ 2d ^(13, 16)
- **Episode Day 2, One:** ⁽¹⁷⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹⁸⁾
 - Crush syndrome and myoglobinuria absent
 - Pain controlled or manageable ⁽¹⁹⁾
 - Tolerating PO ⁽²⁰⁾
 - **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
 - For rhabdomyolysis or crush syndrome use Episode Day 2 at a higher level of care
 - For a condition other than rhabdomyolysis or crush syndrome use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 12h, **All:** ⁽¹⁸⁾
 - **Creatine phosphokinase (CPK), ≥ One:** ⁽⁹⁾
 - Declining and within acceptable limits ⁽²¹⁾
 - ≤ 1000 U/L(17.0 μkat/L)
 - Myoglobinuria absent ⁽⁶⁾
 - **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽²²⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - **Acute kidney injury (AKI) and, ≥ One:**
 - Kidney function improving or within acceptable limits ⁽²¹⁾
 - Outpatient dialysis regimen established
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽²¹⁾
 - **Functional impairment and, One:** ⁽²³⁾
 - **Discharge to home and, One:** ⁽²⁴⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - **Heart failure resolving and, Both:**
 - Dyspnea resolving or at baseline ⁽²⁵⁾
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽²¹⁾
 - Pain controlled or manageable ⁽¹⁹⁾
 - Electrolytes within acceptable limits ⁽²¹⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - **Rhabdomyolysis or crush syndrome and, Both:** ^(4, 5)
 - **Finding, ≥ One:**

- Creatine phosphokinase (CPK) > 1,000 U/L(17.0 μ kat/L) ⁽⁹⁾
- Myoglobinuria ⁽⁶⁾
- Intervention, **All**:
 - Electrolyte monitoring
 - Renal function monitoring
 - IV fluid ^(12, 13)
- Acute kidney injury (AKI) and, **Both**: ⁽²⁶⁾
 - Finding, $\geq$ **One**:
 - Urine output < 0.5 mL/kg/h
 - Creatinine, **One**:
 - ≥ 1.5 x baseline and chronic kidney disease
 - ≥ 1.5 x ULN ⁽²⁷⁾
 - GFR > 25%(0.25) decrease from baseline ⁽²⁵⁾
 - Dialysis initiated
 - Intervention, $\geq$ **One**:
 - Volume expander ≤ 2 d ^(28, 29)
 - Diuretic ≥ 2 x/24h, ≤ 3 d
 - Medication adjustment or discontinuation ≤ 3 d and, $\geq$ **One**: ⁽³⁰⁾
 - Diuretic (includes PO)
 - Nephrotoxic agent (includes PO) ⁽³¹⁾
 - Dialysis (initial) and ≤ 7 d since initiation ⁽³²⁾
 - Dialysis discontinued requiring lab monitoring ≤ 2 d since discontinuation ⁽³³⁾
- Blood glucose ≤ 24 h, **All**: ⁽³⁴⁾
 - Finding, $\geq$ **One**:
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring at least 4x/24h
 - Insulin adjustment and, **One**: ^(34, 35)
 - IV ≥ 2 x/24h
 - SC ≥ 4 x/24h
- Functional impairment (new) requiring rehabilitation initiation ≤ 24 h and, $\geq$ **One**: ^(23, 36)
 - PT evaluation and training
 - OT evaluation and training
 - Speech-language evaluation and training
 - Cognitive evaluation and retraining
 - Swallowing evaluation and retraining
- Heart failure and, **Both**: ⁽³⁷⁾
 - Dyspnea > baseline and, $\geq$ **One**: ⁽²⁵⁾
 - O₂ sat ≤ 91 %(0.91) and < baseline ⁽²⁵⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽³⁸⁾
 - Intervention, $\geq$ **One**: ⁽³⁹⁾
 - Diuretic, **One**: ⁽⁴⁰⁾
 - ≥ 2 x/24h (includes PO)
 - At least 1x/24h and requires daily medication adjustment (includes PO) and, $\geq$ **One**: ⁽³⁰⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽²⁵⁾
 - Creatinine increased from baseline ⁽²⁵⁾
 - Dialysis
- Hyperkalemia, **All**:
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(41, 42)
 - Intervention, $\geq$ **One**:
 - Calcium chloride or gluconate

- Diuretic (includes PO)
- Glucose 50%(0.50) with insulin ⁽¹⁵⁾
- IV fluid ⁽¹²⁾
- Potassium binding agent (includes PO or PR) ⁽⁴³⁾
- Pain uncontrolled and, **Both:**
 - Pain assessment ⁽⁴⁴⁾
 - Requiring change in medication, dose, or frequency ^(30, 45)
 - Post Critical care ≤ 24h
- INTERMEDIATE, ≥ **One:** ⁽⁴⁶⁾
 - Hyperkalemia, **All:**
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, ≥ **One:** ⁽⁴⁷⁾
 - Peaked T waves
 - PVCs > 6/min
 - Intervention, ≥ **One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
- NIPPV ⁽⁴⁸⁾
- Oxygen ≥ 40%(0.40) ≤ 2d ⁽⁴⁹⁾
- CRITICAL, ≥ **One:**
 - Rhabdomyolysis or crush syndrome and, ≥ **One:** ^(4, 5)
 - Dialysis, **One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - CRRT ⁽¹⁴⁾
 - IV fluid resuscitation ≤ 2d ^(13, 16)
 - Acute kidney injury (AKI) and, **Both:** ⁽²⁶⁾
 - Finding, ≥ **One:**
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹¹⁾
 - Widening QRS
 - Sodium, **One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One:**
 - Calcium chloride or gluconate
 - IV fluid resuscitation ≤ 2d ^(13, 16)
 - Diuretic, continuous
 - Dialysis, **One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - CRRT ⁽¹⁴⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** ⁽⁵⁰⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Hemodynamic monitoring, invasive ⁽⁵¹⁾
 - Hyperkalemia, **Both:** ⁽⁴⁷⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
 - Loss of P wave
 - Multifocal PVCs

- Neuromuscular deficit ⁽¹¹⁾
- Widening QRS
- Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Dialysis, **One**:
 - CRRT ⁽¹⁴⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
 - Mechanical ventilation
 - NIPPV and, ≥ **One**: ⁽⁴⁸⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁵²⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3, One**:
 - **ACUTE, One**: ⁽⁸⁾
 - **Early responder**, discharge expected today if clinically stable last 12h, **All**: ⁽¹⁸⁾
 - Creatine phosphokinase (CPK), ≥ **One**: ⁽⁹⁾
 - Declining and within acceptable limits ⁽²¹⁾
 - ≤ 1000 U/L(17.0 µkat/L)
 - Myoglobinuria absent ⁽⁶⁾
 - Complication or comorbidity, **One**:
 - No complication or active comorbidity relevant to this episode of care ⁽²²⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One**:
 - Acute kidney injury (AKI) and, ≥ **One**:
 - Kidney function improving or within acceptable limits ⁽²¹⁾
 - Outpatient dialysis regimen established
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽²¹⁾
 - Functional impairment and, **One**: ⁽²³⁾
 - Discharge to home and, **One**: ⁽²⁴⁾
 - Patient or caregiver able to safely manage impairment
 - Home care services arranged
 - Discharge to inpatient facility for rehabilitative services
 - Heart failure resolving and, **Both**:
 - Dyspnea resolving or at baseline ⁽²⁵⁾
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽²¹⁾
 - Pain controlled or manageable ⁽¹⁹⁾
 - Electrolytes within acceptable limits ⁽²¹⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Rhabdomyolysis or crush syndrome and, **Both**: ^(4, 5)
 - Finding, ≥ **One**:
 - Creatine phosphokinase (CPK) > 1,000 U/L(17.0 µkat/L) ⁽⁹⁾
 - Myoglobinuria ⁽⁶⁾
 - Intervention, **All**:
 - Electrolyte monitoring
 - Renal function monitoring
 - IV fluid ^(12, 13)
 - Acute kidney injury (AKI) and, **Both**: ⁽²⁶⁾
 - Finding, ≥ **One**:
 - Urine output < 0.5 mL/kg/h
 - Creatinine, **One**:
 - ≥ 1.5x baseline and chronic kidney disease
 - ≥ 1.5x ULN ⁽²⁷⁾
 - GFR > 25%(0.25) decrease from baseline ⁽²⁵⁾

- Dialysis initiated
- Intervention, ≥ **One**:
 - Volume expander ≤ 2d ^(28, 29)
 - Diuretic ≥ 2x/24h, ≤ 3d
 - Medication adjustment or discontinuation ≤ 3d and, ≥ **One**: ⁽³⁰⁾
 - Diuretic (includes PO)
 - Nephrotoxic agent (includes PO) ⁽³¹⁾
 - Dialysis (initial) and ≤ 7d since initiation ⁽³²⁾
 - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation ⁽³³⁾
- Blood glucose ≤ 24h, **All**: ⁽³⁴⁾
 - Finding, ≥ **One**:
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring at least 4x/24h
 - Insulin adjustment and, **One**: ^(34, 35)
 - IV ≥ 2x/24h
 - SC ≥ 4x/24h
- Functional impairment (new) requiring rehabilitation initiation ≤ 24h and, ≥ **One**: ^(23, 36)
 - PT evaluation and training
 - OT evaluation and training
 - Speech-language evaluation and training
 - Cognitive evaluation and retraining
 - Swallowing evaluation and retraining
- Heart failure and, **Both**: ⁽³⁷⁾
 - Dyspnea > baseline and, ≥ **One**: ⁽²⁵⁾
 - O₂ sat ≤ 91%(0.91) and < baseline ⁽²⁵⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽³⁸⁾
 - Intervention, ≥ **One**: ⁽³⁹⁾
 - Diuretic, **One**: ⁽⁴⁰⁾
 - ≥ 2x/24h (includes PO)
 - At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One**: ⁽³⁰⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽²⁵⁾
 - Creatinine increased from baseline ⁽²⁵⁾
 - Dialysis
- Hyperkalemia, **All**:
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(41, 42)
 - Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
 - IV fluid ⁽¹²⁾
 - Potassium binding agent (includes PO or PR) ⁽⁴³⁾
- Pain uncontrolled and, **Both**:
 - Pain assessment ⁽⁴⁴⁾
 - Requiring change in medication, dose, or frequency ^(30, 45)
 - Post Critical care ≤ 24h
- INTERMEDIATE, ≥ **One**:
 - Hyperkalemia, **All**:
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, ≥ **One**: ⁽⁴⁷⁾
 - Peaked T waves

- PVCs > 6/min
- Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
- NIPPV ⁽⁴⁸⁾
- Oxygen ≥ 40%(0.40) ≤ 2d ⁽⁴⁹⁾
- **CRITICAL**, ≥ **One**:
 - Rhabdomyolysis or crush syndrome and, ≥ **One**: ^(4, 5)
 - Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - CRRT ⁽¹⁴⁾
 - IV fluid resuscitation ≤ 2d ^(13, 16)
 - Acute kidney injury (AKI) and, **Both**: ⁽²⁶⁾
 - Finding, ≥ **One**:
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹¹⁾
 - Widening QRS
 - Sodium, **One**:
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - IV fluid resuscitation ≤ 2d ^(13, 16)
 - Diuretic, continuous
 - Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - CRRT ⁽¹⁴⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One**: ⁽⁵⁰⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Hemodynamic monitoring, invasive ⁽⁵¹⁾
 - Hyperkalemia, **Both**: ⁽⁴⁷⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹¹⁾
 - Widening QRS
 - Intervention, ≥ **One**:
 - Calcium chloride or gluconate
 - Dialysis, **One**:
 - CRRT ⁽¹⁴⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
 - Mechanical ventilation
 - NIPPV and, ≥ **One**: ⁽⁴⁸⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁵²⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● Episode Day 4, One:

● ACUTE, One: ⁽⁸⁾● Responder, discharge expected today if clinically stable last 12h, All: ⁽¹⁸⁾● Creatine phosphokinase (CPK), ≥ One: ⁽⁹⁾

- Declining and within acceptable limits ⁽²¹⁾
- ≤ 1000 U/L(17.0 μkat/L)

● Myoglobinuria absent ⁽⁶⁾

● Complication or comorbidity, One:

- No complication or active comorbidity relevant to this episode of care ⁽²²⁾

● Complication or comorbidity and clinically stable for discharge, ≥ One:

● Acute kidney injury (AKI) and, ≥ One:

- Kidney function improving or within acceptable limits ⁽²¹⁾
- Outpatient dialysis regimen established

● Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽²¹⁾● Functional impairment and, One: ⁽²³⁾● Discharge to home and, One: ⁽²⁴⁾

- Patient or caregiver able to safely manage impairment
- Home care services arranged
- Discharge to inpatient facility for rehabilitative services

● Heart failure resolving and, Both:

- Dyspnea resolving or at baseline ⁽²⁵⁾
- O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽²¹⁾
- Pain controlled or manageable ⁽¹⁹⁾
- Electrolytes within acceptable limits ⁽²¹⁾

● Partial responder, not clinically stable for discharge and requires continued stay, ≥ One:

● Rhabdomyolysis or crush syndrome and, Both: ^(4, 5)

● Finding, ≥ One:

- Creatine phosphokinase (CPK) > 1,000 U/L(17.0 μkat/L) ⁽⁹⁾
- Myoglobinuria ⁽⁶⁾

● Intervention, All:

- Electrolyte monitoring
- Renal function monitoring
- IV fluid ^(12, 13)

● Acute kidney injury (AKI) and, Both: ⁽²⁶⁾

● Finding, ≥ One:

- Urine output < 0.5 mL/kg/h

● Creatinine, One:

- ≥ 1.5x baseline and chronic kidney disease
- ≥ 1.5x ULN ⁽²⁷⁾
- GFR > 25%(0.25) decrease from baseline ⁽²⁵⁾
- Dialysis initiated

● Intervention, ≥ One:

- Volume expander ≤ 2d ^(28, 29)
- Diuretic ≥ 2x/24h, ≤ 3d
- Medication adjustment or discontinuation ≤ 3d and, ≥ One: ⁽³⁰⁾
 - Diuretic (includes PO)
 - Nephrotoxic agent (includes PO) ⁽³¹⁾
- Dialysis (initial) and ≤ 7d since initiation ⁽³²⁾
- Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation ⁽³³⁾

● Blood glucose ≤ 24h, All: ⁽³⁴⁾

● Finding, ≥ One:

- > 250 mg/dL(13.9 mmol/L)
- < 70 mg/dL(3.9 mmol/L)

- Blood glucose monitoring at least 4x/24h
- Insulin adjustment and, **One:** ^(34, 35)
 - IV $\geq 2x/24h$
 - SC $\geq 4x/24h$
- Functional impairment (new) requiring rehabilitation initiation $\leq 24h$ and, $\geq$ **One:** ^(23, 36)
 - PT evaluation and training
 - OT evaluation and training
 - Speech-language evaluation and training
 - Cognitive evaluation and retraining
 - Swallowing evaluation and retraining
- Heart failure and, **Both:** ⁽³⁷⁾
 - Dyspnea $>$ baseline and, $\geq$ **One:** ⁽²⁵⁾
 - O_2 sat $\leq 91\%(0.91)$ and $<$ baseline ⁽²⁵⁾
 - Worsening edema
 - Heart failure confirmed by imaging ⁽³⁸⁾
 - Intervention, $\geq$ **One:** ⁽³⁹⁾
 - Diuretic, **One:** ⁽⁴⁰⁾
 - $\geq 2x/24h$ (includes PO)
 - At least $1x/24h$ and requires daily medication adjustment (includes PO) and, $\geq$ **One:** ⁽³⁰⁾
 - Blood urea nitrogen (BUN) increased from baseline ⁽²⁵⁾
 - Creatinine increased from baseline ⁽²⁵⁾
 - Dialysis
- Hyperkalemia, **All:**
 - Potassium > 5.5 mEq/L(5.5 mmol/L)
 - No electrocardiogram (ECG) changes ^(41, 42)
 - Intervention, $\geq$ **One:**
 - Calcium chloride or gluconate
 - Diuretic (includes PO)
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
 - IV fluid ⁽¹²⁾
 - Potassium binding agent (includes PO or PR) ⁽⁴³⁾
- Pain uncontrolled and, **Both:**
 - Pain assessment ⁽⁴⁴⁾
 - Requiring change in medication, dose, or frequency ^(30, 45)
 - Post Critical care $\leq 24h$
- INTERMEDIATE, $\geq$ **One:**
 - Hyperkalemia, **All:**
 - Potassium ≥ 5.6 mEq/L(5.6 mmol/L)
 - Electrocardiogram (ECG) changes, $\geq$ **One:** ⁽⁴⁷⁾
 - Peaked T waves
 - PVCs $> 6/min$
 - Intervention, $\geq$ **One:**
 - Calcium chloride or gluconate
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
- NIPPV ⁽⁴⁸⁾
- Oxygen $\geq 40\%(0.40) \leq 2d$ ⁽⁴⁹⁾
- CRITICAL, $\geq$ **One:**
 - Rhabdomyolysis or crush syndrome and, $\geq$ **One:** ^(4, 5)
 - Dialysis, **One:**
 - Hemodialysis $\leq 7d$ since initiation
 - Peritoneal dialysis $\leq 7d$ since initiation
 - CRRT ⁽¹⁴⁾
 - IV fluid resuscitation $\leq 2d$ ^(13, 16)

- Acute kidney injury (AKI) and, **Both:** ⁽²⁶⁾
 - Finding, **≥ One:**
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, **≥ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹¹⁾
 - Widening QRS
 - Sodium, **One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, **≥ One:**
 - Calcium chloride or gluconate
 - IV fluid resuscitation ≤ 2d ^(13, 16)
 - Diuretic, continuous
 - Dialysis, **One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
- CRRT ⁽¹⁴⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **≥ One:** ⁽⁵⁰⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
- Hemodynamic monitoring, invasive ⁽⁵¹⁾
- Hyperkalemia, **Both:** ⁽⁴⁷⁾
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, **≥ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽¹¹⁾
 - Widening QRS
 - Intervention, **≥ One:**
 - Calcium chloride or gluconate
 - Dialysis, **One:**
 - CRRT ⁽¹⁴⁾
 - Dialysis, hemodialysis or peritoneal
 - Glucose 50%(0.50) with insulin ⁽¹⁵⁾
- Mechanical ventilation
- NIPPV and, **≥ One:** ⁽⁴⁸⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁵²⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 5, **One:**
 - ACUTE, **One:** ⁽⁸⁾
 - Responder, discharge expected today if clinically stable last 12h, **All:** ⁽¹⁸⁾
 - Creatine phosphokinase (CPK), **≥ One:** ⁽⁹⁾
 - Declining and within acceptable limits ⁽²¹⁾
 - ≤ 1000 U/L(17.0 μkat/L)
 - Myoglobinuria absent ⁽⁶⁾
 - Complication or comorbidity, **One:**
 - No complication or active comorbidity relevant to this episode of care ⁽²²⁾
 - Complication or comorbidity and clinically stable for discharge, **≥ One:**
 - Acute kidney injury (AKI) and, **≥ One:**
 - Kidney function improving or within acceptable limits ⁽²¹⁾
 - Outpatient dialysis regimen established

– Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽²¹⁾

● **Functional impairment and, One:** ⁽²³⁾

● **Discharge to home and, One:** ⁽²⁴⁾

- Patient or caregiver able to safely manage impairment
- Home care services arranged

– Discharge to inpatient facility for rehabilitative services

● **Heart failure resolving and, Both:**

- Dyspnea resolving or at baseline ⁽²⁵⁾
- O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽²¹⁾

– Pain controlled or manageable ⁽¹⁹⁾

– Electrolytes within acceptable limits ⁽²¹⁾

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

● **INTERMEDIATE, One:**

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

● **CRITICAL, One:**

● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**

- **Use Extended Stay subset**
- Refer for secondary review

Discharge, Level of Care, One: ⁽⁵³⁾

● **HOME, Both:**

● **Level of care appropriateness, Both:**

- Home environment safe and accessible ⁽⁵⁴⁾
- Patient or caregiver demonstrates ability to manage care

● **Discharge planning, All:**

- Provide comprehensive written discharge and teaching instructions ⁽⁵⁵⁾
- Medication reconciliation ⁽⁵⁶⁾
- Follow-up care planned ⁽⁵⁷⁾
- Identify and address transportation needs

● **HOME CARE, Both:**

● **Level of care appropriateness, All:**

- Home environment safe and accessible ⁽⁵⁴⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(58, 59)

● **Skilled services, ≥ One:** ⁽⁶⁰⁾

- Skilled nursing ⁽⁶¹⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽⁶²⁾

● **Discharge planning, All:**

- Provide comprehensive written discharge and teaching instructions ⁽⁵⁵⁾
- Medication reconciliation ⁽⁵⁶⁾
- Follow-up care planned ⁽⁵⁷⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**

● **Level of care appropriateness, One:**

- Support systems available ⁽⁶³⁾

- Skilled treatment, ≥ **One**:
 - Clinical assessment ⁽⁶⁴⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation
- **SKILLED MEDICAL OR THERAPY, Both**:
 - Level of care appropriateness, **All**:
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One**:
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁶¹⁾
 - Therapy, **All**:
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - Functional impairments requiring at least supervision ⁽⁶⁷⁾
 - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
 - Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁵⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both**:
 - Level of care appropriateness, **All**:
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One**:
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁶¹⁾
 - Therapy, **All**:
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁵⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both**:
 - Level of care appropriateness, **All**:
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - Services, **All**:
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
 - Complete prior to facility transfer, **All**:
 - Medication reconciliation ⁽⁵⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both**:
 - Level of care appropriateness, **Both**:
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Services, **All**:
 - Goal directed therapy with rehabilitation potential ^(65, 66)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁶⁷⁾

– Able to tolerate $\geq 15\text{h/wk}$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁶⁸⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁵⁶⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

● Level of care appropriateness, **Both:**

- Treatment precluded in a lower level

● In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**

- Primary condition or illness and treatment of active comorbid condition(s) ⁽⁶⁹⁾
- Ventilator dependent $\geq 6\text{h/d}$ and weaning planned ^(70, 71)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽⁷²⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁵⁶⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
557 TENDONITIS, MYOSITIS AND BURSITIS WITH MCC	n/a	5.1	CMS GMLOS
558 TENDONITIS, MYOSITIS AND BURSITIS WITHOUT MCC	n/a	3.1	CMS GMLOS
564 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DIAGNOSES WITH MCC	n/a	5	CMS GMLOS
565 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DIAGNOSES WITH CC	n/a	3.3	CMS GMLOS
566 OTHER MUSCULOSKELETAL SYSTEM AND CONNECTIVE TISSUE DIAGNOSES WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
CRUSH SYNDROME	16%	4.9	InterQual
RHABDOMYOLYSIS	22%	4.1	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

Introduction:

Rhabdomyolysis is the degeneration (lysis) of skeletal muscle due to trauma, severe compression (e.g., compartment syndrome, crush syndrome), drugs, toxins, or infections. Lysis of muscle cells releases toxic intracellular components into the systemic circulation which may lead to electrolyte disturbances, hypovolemia, metabolic acidosis, coagulation defects, or acute kidney injury. Excessive physical exertion as well as direct muscle trauma can provoke rhabdomyolysis. Rhabdomyolysis is often observed in patients following traumatic injuries from motor vehicle accidents, industrial accidents, deep tissue burns, or severe physical abuse. Additionally, rhabdomyolysis may be observed in body-builders, marathon runners, individuals with substance use disorder, delirium tremens, or following protracted tonic-clonic seizures (Torres et al., The Ochsner Journal 2015, 15: 58-69).

5:

Crush injury involves damage to striated muscle. Crush syndrome is the term used to describe complications resulting from muscle damage. Crush syndrome results in a characteristic syndrome of traumatic rhabdomyolysis, myoglobinuria, and acute renal failure. Renal and cardiac complications are dependent upon the magnitude of the pressure and the size of the compressed muscle group. Entrapment with muscle trauma can lead to compartment syndrome or rhabdomyolysis. Rhabdomyolysis under these conditions is potentially fatal (Gonzalez, Crit Care Med 2005; 33(1 Suppl): S34-41; Cervellin et al., Clin Chem Lab Med 2010, 48: 749-56). The Observation level of care is appropriate for patients with crush injury when there is concern for progression to a more severe state.

6:

Myoglobinuria refers to an excess amount of myoglobin found in the urine due to muscle trauma, burns, etc. It may also be diagnosed from a urine analysis that is heme-positive but has no red blood cells.

7:

Prolonged compression of muscle is usually defined as compression lasting more than 4-6 hours. Under certain conditions, damage may occur in as little as one hour and is sufficient to meet criteria (Gonzalez, Crit Care Med 2005; 33(1 Suppl): S34-41).

8:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

9:

With rhabdomyolysis, creatine phosphokinase (CPK) typically rises in the first 12 hours following injury, peaks at 3 days, and normalizes at around 5 days. There is a direct correlation between the degree of CPK elevation and the degree of skeletal muscle injury (Parekh et al., Emerg Med Pract 2012, 14: 1-15).

10:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment,

assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

11:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

12:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

13:

Patients with rhabdomyolysis require aggressive fluid resuscitation to avoid myoglobin-induced acute renal failure. Studies recommend an intravenous (IV) infusion of 0.9% saline at a rate of 400mL/hr. The patient's urine output must be closely monitored with goal of 200mL/hr until the creatine phosphokinase (CPK), blood urea nitrogen (BUN), creatinine, and urine pH normalize (Parekh et al., Emerg Med Pract 2012, 14: 1-15).

14:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

15:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

16:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children:

Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

17:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

18:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

19:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

20:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

21:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

22:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

23:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

24:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

25:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

26:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.

AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.

AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

27:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

28:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume

depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

29:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Guideline Updates, In: *Acute kidney injury: prevention, detection and management*. 2019; Pistolesi et al., *J Nephrol* 2018, 31: 797-812; Joannidis et al., *Intensive Care Med* 2017, 43: 730-49; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii-138).

30:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

31:

Nephrotoxic agents include nonsteroidal anti-inflammatory drugs (NSAIDs), aminoglycosides, amphotericin B, high doses of loop diuretics, and anti-viral agents or other drugs (e.g., acyclovir and foscarnet) (Guideline Updates, In: *Acute kidney injury: prevention, detection and management*. 2019; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii-138).

32:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

33:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

34:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, *Diabetes Care* 2024, 47: S1-S314).

35:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

36:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

37:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor

(ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

38:

Bilateral pulmonary infiltrates seen on chest x-ray are consistent with the diagnosis of left-sided heart failure.

39:

In patients hospitalized with acute heart failure (HF) due to an increased risk of atrial and ventricular arrhythmias, continuous cardiac monitoring is recommended. Monitoring may be discontinued when there is no evidence of a hemodynamically significant arrhythmia and the symptoms of acute HF have improved (Sandau et al., Circulation 2017, 136: e273-e344).

40:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

41:

In patients with no underlying heart disease, occasional, nonconsecutive premature ventricular contractions (PVCs) less than 6/min are common and not considered a concerning finding. Three or more consecutive PVCs at a rate of 100 bpm or more that terminate spontaneously within 30 seconds are considered nonsustained ventricular tachycardia (Marine, Card Electrophysiol Clin 2016, 8: 525-43; Buxton et al., Circulation 2006, 114: 2534-70).

42:

Following an initial 12-lead electrocardiogram (ECG) demonstrating no abnormal findings in the setting of hyperkalemia or hypokalemia, continued monitoring may be accomplished by telemetry (Bridgeman et al., Nephrol Dial Transplant 2019, 34: iii45-iii50).

43:

Examples of potassium binding agents include but are not limited to:

- Sodium polystyrene sulfonate (Kayexalate® or Kionex®)
- Sodium zirconium cyclosilicate (Lokelma®)
- Patiromer (Veltassa®)

44:

Pain assessment includes quality, intensity, location, frequency, duration, and exacerbating or alleviating factors. Uncontrolled pain should be monitored, assessed, and documented. Appropriate medication adjustment and other interventions should be taken to reduce pain.

45:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug

monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription

- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

46:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

47:

Hyperkalemia may cause electrocardiogram (ECG) changes such as a tall, narrow, T wave and a widening of the QRS complex (Sandau et al., Circulation 2017, 136: e273-e344).

48:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

49:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula**Room air** 21%(0.21)**1L** 24%(0.24) **4L** 36%(0.36)**2L** 28%(0.28) **5L** 40%(0.40)**3L** 32%(0.32) **6L** 44%(0.44)**Simple oxygen masks**

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

50:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

51:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

52:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

53:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

54:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack

of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

55:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

56:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

57:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

58:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk

- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

59:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

60:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

61:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

62:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care,

or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

63:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

64:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

65:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

66:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

67:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or

there are safety (risk) considerations

- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

68:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

69:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

70:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

71:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

72:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid

Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.